

Pixel-wise spectral coding and direct resonance-shift estimation

The Stage 1 forward model now feeds a camera instead of two detectors. Each pixel is a measurement with its own spectral weighting, and the resonance shift $\Delta\lambda_r$ is estimated directly from the most informative pixels, then converted to a refractive-index change Δn . The DOE is still the Stage 1 linear grating: no inverse design yet.

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0.073 nm

Cramér–Rao bound on $\Delta\lambda_r$, all 2304 pixels (Monte Carlo 0.074 nm)

3.67×10^{-4}

$\sigma_{\Delta n}$ in RIU, for $S_n = 200$ nm/RIU (assumed)

0.258 nm

bound with the best 16 pixels; 256 pixels are within 15 % of all

-0.62 ... +0.69 nm

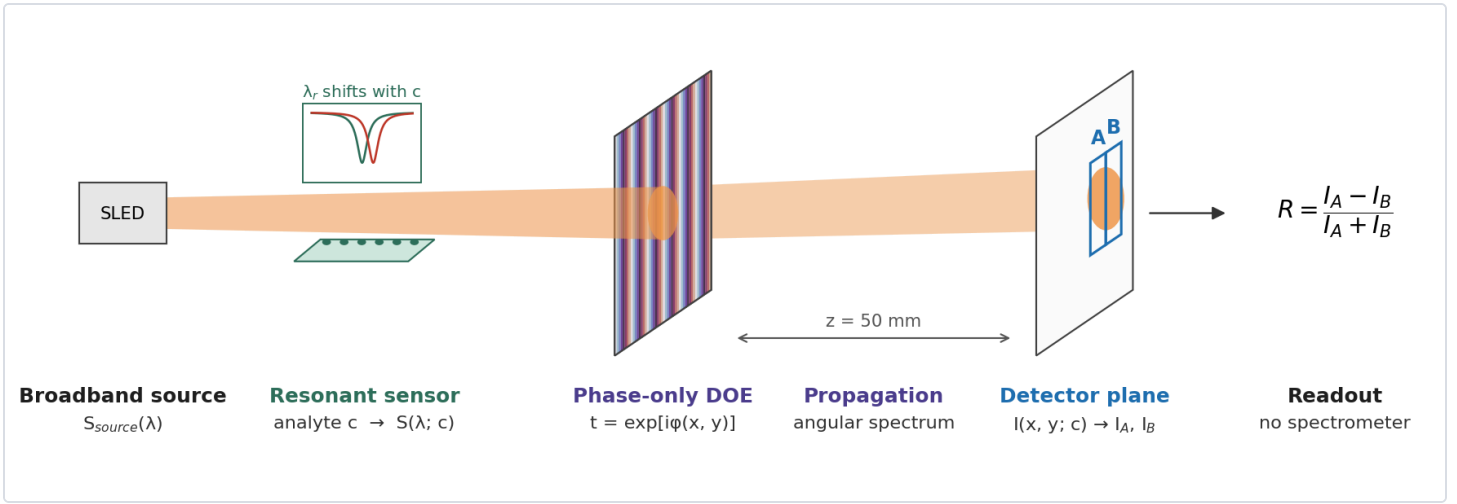
linear range: linearization bias below the all-pixel noise ($\gamma = 2.5$ nm)

What changed from Stage 1

Stage 1 reduced the detector image to two regions A and B and a ratio R. Embodiment 2 keeps the individual camera pixels as separate measurements:



Kept unchanged from Stage 1: the Lorentzian sensor ($\lambda_{r,0} = 1550$ nm, $\gamma = 2.5$ nm, $A = 0.8$) with its 20 nm SLED source, the 150-sample wavelength grid (1520.0–1585.0 nm), the band-limited angular-spectrum propagation of every wavelength, incoherent summation, and the 160 μm linear phase grating at $z = 50$ mm. The Stage 1 scripts, results and checks are untouched; the two-region readout is kept only as a comparison.



Stage 1 architecture, for reference. Embodiment 2 replaces the A/B split and the ratio R with the full camera vector and a statistical estimator.

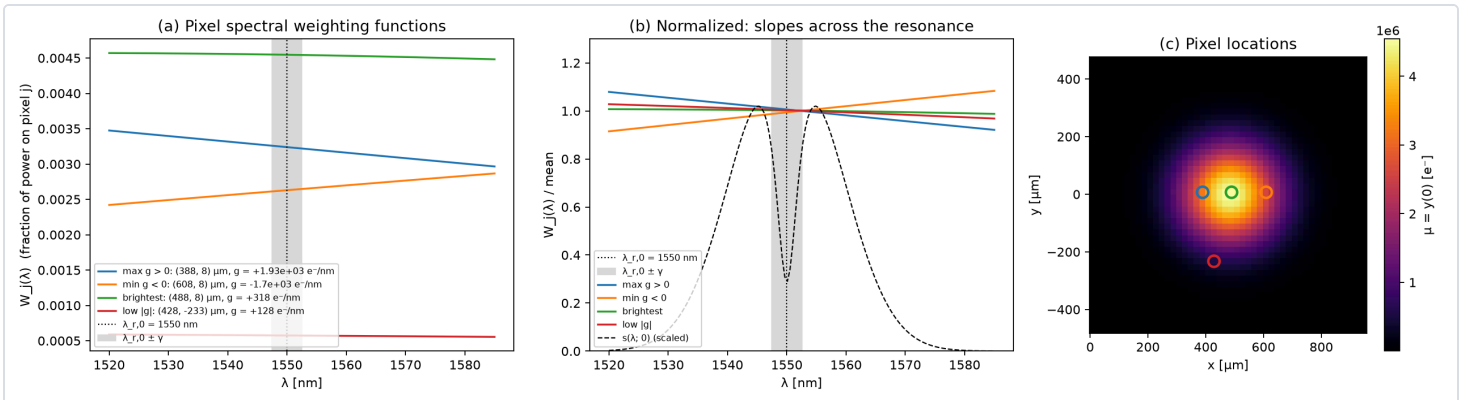
1 · Pixel spectral weighting functions

For each wavelength, a unit-power Gaussian beam is propagated through the grating and the power on each camera pixel is recorded. Row j of the matrix W is the spectral weighting $W_j(\lambda)$ of pixel j :

$$W_{ji} = \iint_{\Omega_j} |E_i(x, y)|^2 dx dy, \quad \iint |E_{in}|^2 = 1 \quad (\text{Eq. 4})$$

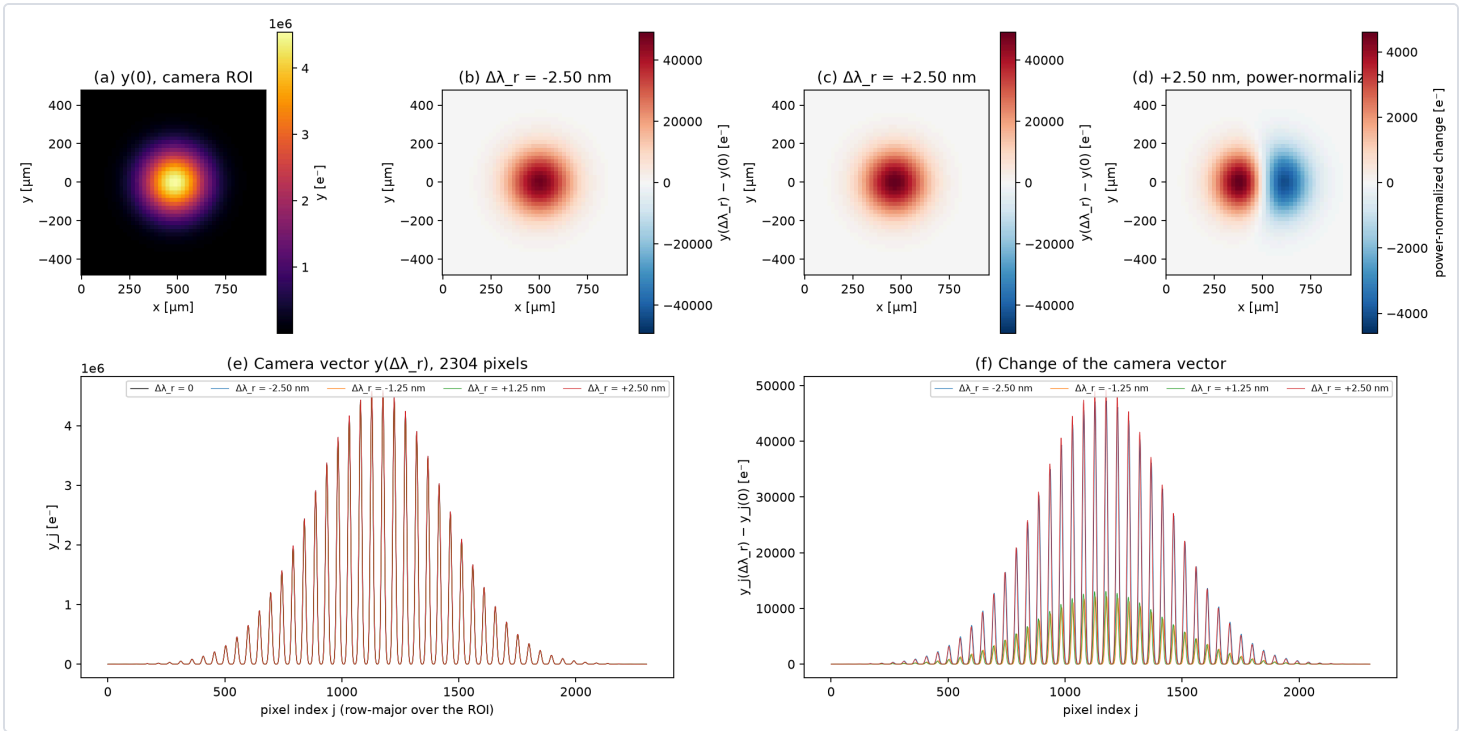
$$y_j(\Delta\lambda_r) = \sum_i W_{ji} s(\lambda_i; \Delta\lambda_r) \Delta\lambda \quad \Leftrightarrow \quad y = W s \quad (\text{Eq. 5})$$

The camera has $20 \mu\text{m}$ pixels (4×4 simulation cells). A 48×48 pixel region of interest around the $+1$ order holds 99.99 % of the light on the camera. W is computed once (3 s); every new shift is then one matrix–vector product.



Deliverable 1. (a) $W_j(\lambda)$ for the pixels with the largest positive and negative g_j , the brightest pixel and a low- $|g|$ pixel; resonance $\lambda_{r,0} \pm \gamma$ shaded. (b) The same, normalized: the grating gives every pixel an almost linear spectral slope, positive on one side of the spot and negative on the other. (c) Pixel locations on $y(0)$.

2 • The camera vector for known shifts



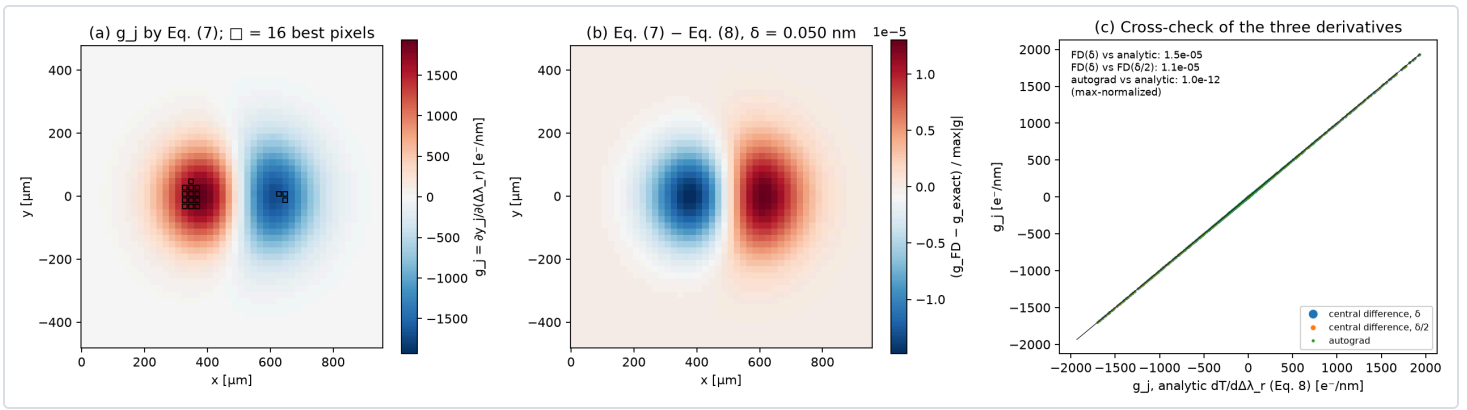
Deliverable 2. (a) $y(0)$ on the ROI, peak $4.54e+06$ e^- per measurement ($4.54e+04$ e^- per frame). (b, c) $y(\Delta\lambda_r) - y(0)$ for ± 2.50 nm shifts: both shifts brighten the spot, because the dip moves off the centre of the source spectrum. (d) With the total power removed, the spectral coding appears as a left–right redistribution. (e, f) The complete 2304-pixel vector $y(\Delta\lambda_r)$ and its change, row-major over the ROI.

3 • Resonance-shift sensitivity

$$g_j = \partial y_j / \partial (\Delta\lambda_r) |_0 \approx [y_j(+\delta) - y_j(-\delta)] / 2\delta, \quad \delta = 0.02\gamma = 0.050 \text{ nm} \quad (\text{Eq. 7})$$

$$g = W \partial s / \partial (\Delta\lambda_r), \quad \partial s / \partial (\Delta\lambda_r) = s_{\text{src}} \partial T / \partial (\Delta\lambda_r) \Delta\lambda \quad (\text{Eq. 8})$$

Three independent derivatives agree: the central difference changes by $1.1e-05$ (max-normalized) when δ is halved and matches the analytic derivative of the Lorentzian to $1.5e-05$, and forward-mode autograd through the Stage 1 sensor model matches the analytic result to $1e-12$.



Deliverable 3. (a) Signed g_j [e^-/nm]: an antisymmetric dipole across the +1-order spot. The 16 highest-information pixels are marked. (b) Eq. (7) – Eq. (8): an $O(\delta^2)$ residual at the 10^{-5} level. (c) Finite difference (δ and $\delta/2$) and autograd against the analytic derivative.

4 • Estimator, noise and linear range

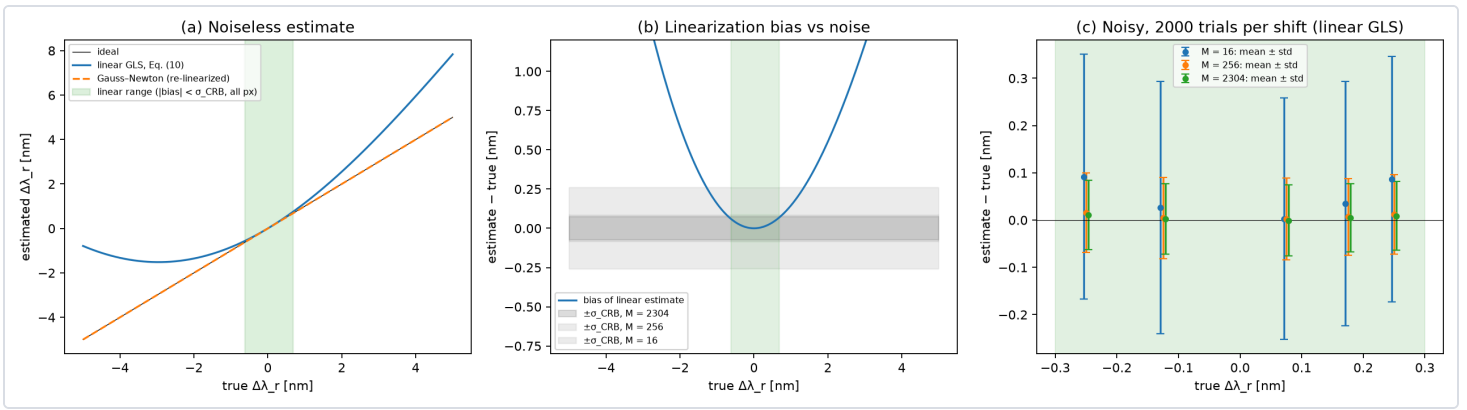
Pixels have independent shot and read noise. The measurement sums 100 frames of 1×10^7 photoelectrons, each with 100 e^- read noise, so the brightest pixel stays near $4.5\text{e}+04 \text{ e}^-$ per frame, below a typical InGaAs full well. The shift is estimated by generalized least squares:

$$\sigma_j^2 = y_j + \sigma_{\text{read}}^2 \quad (\text{Eq. 9})$$

$$\Delta\lambda_r = g^T \Sigma^{-1} (y_{\text{meas}} - \mu) / g^T \Sigma^{-1} g, \quad \text{Var}(\Delta\lambda_r) = 1/F, \quad F = \sum_j g_j^2 / \sigma_j^2 \quad (\text{Eq. 10, 12})$$

$$\Delta\hat{n} = \Delta\lambda_r / S_n, \quad \sigma_{\Delta n} = \sigma_{\Delta\lambda} / S_n \quad (\text{Eq. 12})$$

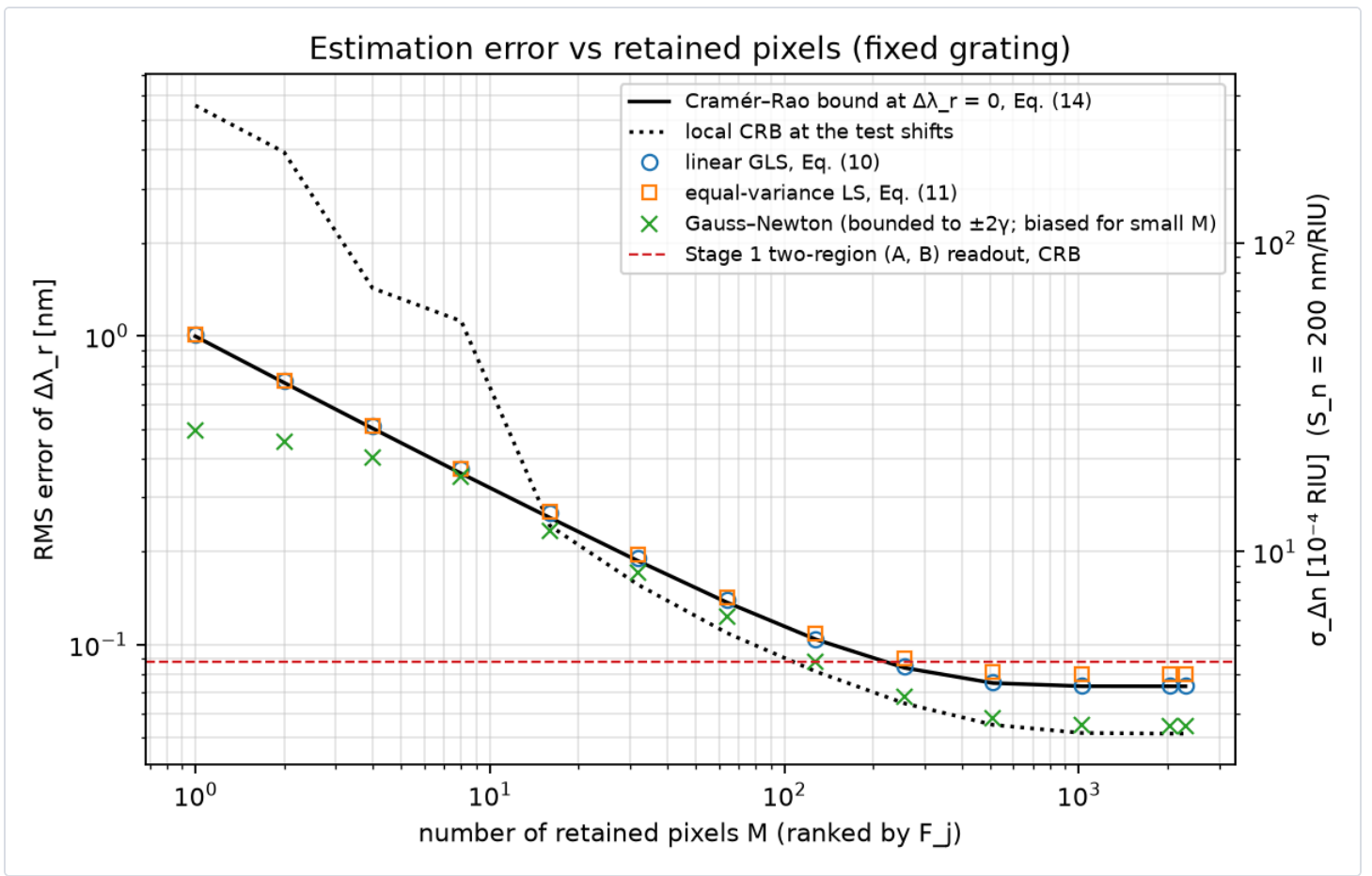
The linear model is exact only near $\Delta\lambda_r = 0$. Its bias grows roughly quadratically (9.4 pm at 0.1 γ) and exceeds the all-pixel noise outside $-0.625 \dots +0.688 \text{ nm}$. With only 16 pixels the noise is larger, so the usable range widens to $-1.25 \dots +1.31 \text{ nm}$. Re-linearizing about the current estimate (Gauss–Newton, 10 iterations) removes the bias to $3\text{e}-13 \text{ nm}$ over $\pm 2\gamma$.



Deliverable 4. (a) Noiseless estimate: linear GLS bends away from the ideal line; Gauss–Newton lies on it. Green: linear range. (b) Linearization bias against $\pm\sigma_{\text{CRB}}$ for 16, 256 and all pixels. (c) Noisy estimates at five test shifts that differ from the calibration points $\pm\delta$ (mean $\pm$ standard deviation over 2000 realizations each).

5 • How many pixels are needed

Pixels are ranked by their Fisher information $F_j = g_j^2/\sigma_j^2$, and the estimator is applied to the best M . The Monte Carlo uses 2000 noise realizations at each of 5 test shifts. Its RMS error matches the exact error of the linear estimator (bias² plus variance) within 1.0 % and stays within 3.4 % of the Cramér–Rao bound for every M .



Deliverable 5. RMS error of $\Delta\lambda_r$ against M. Right axis: $\sigma_{\Delta n}$. The dashed red line is the bound for the Stage 1 two-region readout (A, B) built from the same pixels and photons.

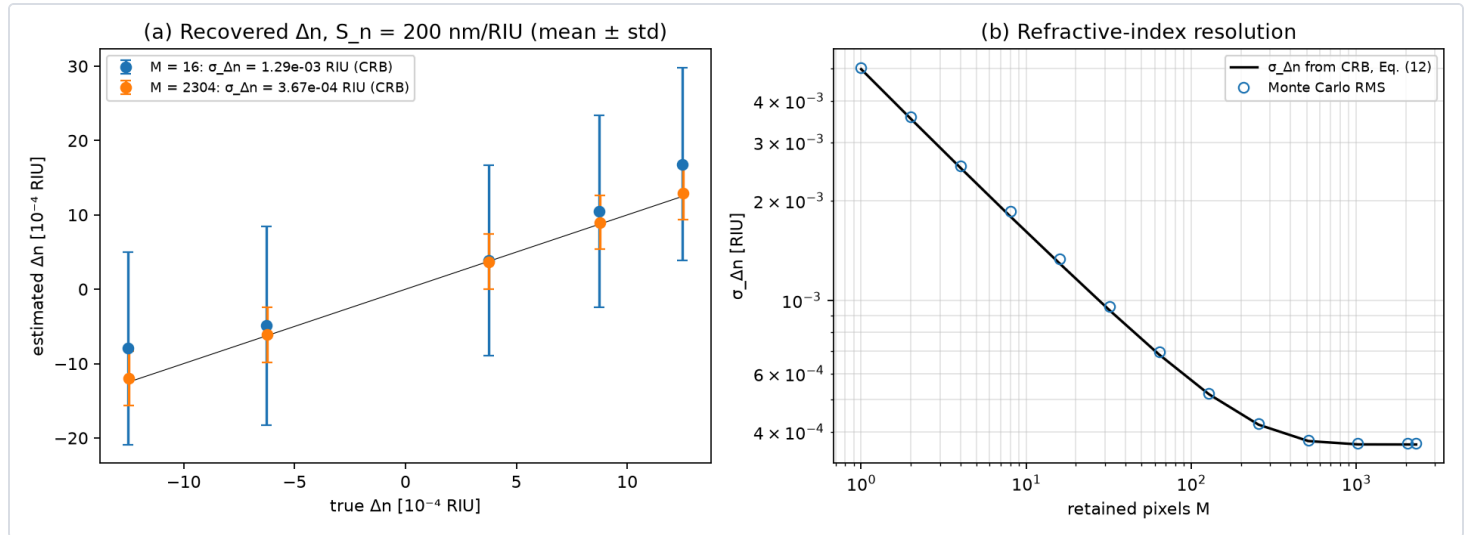
M	CRB at 0 [nm]	RMS GLS [nm]	predicted [nm]	RMS eq-var LS [nm]	local CRB [nm]	RMS Gauss– Newton [nm]	$\sigma_{\Delta n}$ [RIU]
1	0.9967	1.0065	1.0010	1.0065	5.5767	0.4940	4.98e-03
2	0.7058	0.7154	0.7117	0.7154	3.9387	0.4546	3.53e-03
4	0.5017	0.5096	0.5112	0.5108	1.4271	0.4030	2.51e-03
8	0.3583	0.3702	0.3708	0.3723	1.1165	0.3503	1.79e-03
16	0.2577	0.2666	0.2649	0.2692	0.2427	0.2330	1.29e-03
32	0.1864	0.1914	0.1913	0.1957	0.1560	0.1707	9.32e-04
64	0.1369	0.1393	0.1379	0.1428	0.1090	0.1228	6.85e-04
128	0.1040	0.1044	0.1044	0.1085	0.0818	0.0877	5.20e-04
256	0.0841	0.0850	0.0846	0.0906	0.0645	0.0676	4.21e-04
512	0.0751	0.0755	0.0754	0.0817	0.0550	0.0578	3.75e-04
1024	0.0734	0.0739	0.0736	0.0804	0.0517	0.0548	3.67e-04
2048	0.0733	0.0738	0.0736	0.0803	0.0515	0.0545	3.67e-04
2304	0.0733	0.0738	0.0736	0.0803	0.0515	0.0545	3.67e-04

- A few pixels already carry most of the information: M = 16 gives 0.258 nm, M = 256 is within 15 % and M = 512 within 2.4 % of all 2304 pixels.
- Weighting by the shot-plus-read variance matters: equal-variance least squares (Eq. 11) is 8.8 % worse with all pixels.

- With this grating, the pixel-wise estimator improves on the Stage 1 two-region readout only by a factor 1.20 (0.088 $\rightarrow$ 0.073 nm). The grating encodes the shift as a dipole, a centroid move that a two-detector split already captures. Larger gains need a DOE that spreads spectral information over more pixels, which is the purpose of the inverse design.
- For $M \geq 16$, Gauss–Newton falls below the $\Delta\lambda_r = 0$ bound but stays above the local bound at the test shifts (section 7). For $M \leq 8$ it is clipped to $\pm 2\gamma$ and biased, so the bound does not apply.

6 • Refractive-index change

With the assumed bulk sensitivity $S_n = 200$ nm/RIU, all pixels give $\sigma_{\Delta n} = 3.67 \times 10^{-4}$ RIU, and the best 16 give 1.29×10^{-3} RIU. Because Eq. (12) is a division by S_n , these numbers scale directly with the value of S_n of the real resonator.

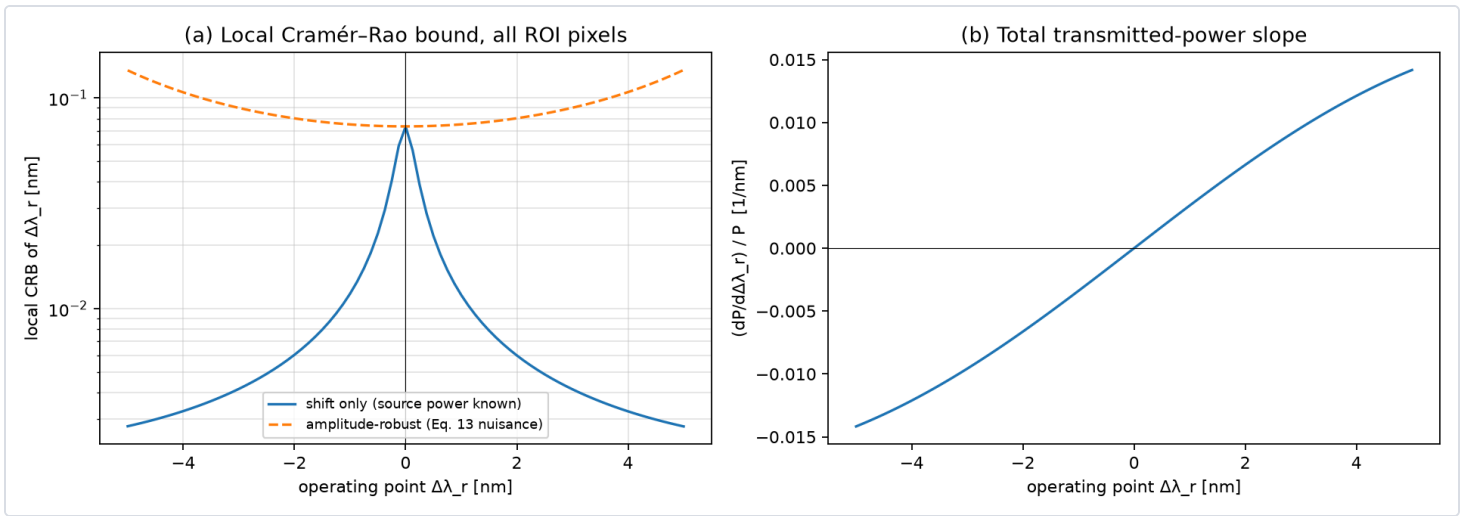


Deliverable 6. (a) Recovered Δn at the five test shifts (mean $\pm$ std) for 16 and all pixels. (b) $\sigma_{\Delta n}$ against M : Eq. (12) applied to the Cramér–Rao bound, and Monte Carlo RMS.

7 • Where the information comes from

The source spectrum is centred on the resonance, so the total transmitted power is stationary at $\Delta\lambda_r = 0$. There, all the information comes from how the spectrum is redistributed across the pixels. Away from zero the total power changes with the shift and adds a strong intensity channel: the local bound drops from 0.073 nm at 0 to 0.0049 nm at $+\gamma$.

That intensity channel is exactly what a drifting source corrupts. With the source amplitude estimated jointly (Eq. 13) the bound at $+\gamma$ is 0.084 nm, close to the $\Delta\lambda_r = 0$ value. A 1 % power change read by Eq. (10) gives a false shift of 0.89 nm (12 σ); the joint fit removes it ($2e-13$ nm) for almost no cost at the operating point (0.0733 $\rightarrow$ 0.0734 nm).



Supporting analysis. (a) Local bound with the source power known (shift only) and with the amplitude as a nuisance parameter. (b) Relative slope of the total transmitted power.

Validation

Result	Check	Value
PASS	W: power per wavelength	<code>max sum_j W_ji - (1 - band-limit loss) = 1.1e-16 (limit 1e-12)</code>
PASS	y = W s equals the Stage 1 broadband image	<code>max rel. difference 4.1e-16 (limit 1e-12)</code>
PASS	g: central difference converged in delta	<code>delta vs delta/2: 1.1e-05 (limit 1e-4)</code>
PASS	g: Eq. (7) vs Eq. (8)	<code>FD vs analytic 1.5e-05 (limit 1e-4), autograd vs analytic 1.0e-12 (limit 1e-10)</code>
PASS	Noise model: variance = y + sigma_read^2	<code>variance error 1.9e-02 (limit 5e-2), mean within 2.4 standard errors (limit 5)</code>
PASS	Gauss-Newton removes the linearization bias	<code>max error 3.4e-13 nm over +-2 gamma (limit 1e-9)</code>
PASS	Monte Carlo matches the exact linear-estimator error	<code>max RMS_MC / RMS_predicted - 1 = 0.010 over all M (limit 0.02, 10000 samples per M)</code>
PASS	Monte Carlo RMS reaches the Cramer-Rao bound	<code>max RMS/CRB - 1 = 0.034 over all M (limit 0.05); M = 2304: RMS 0.0738 nm, CRB 0.0733 nm</code>
PASS	Joint amplitude fit removes a 1 % power drift	<code>drift-induced shift error 0.891 nm (Eq. 10) -> 1.9e-13 nm (Eq. 13); CRB cost 0.0733 -> 0.0734 nm</code>

9 of 9 Embodiment 2 checks pass. The Stage 1 validation (7 checks) and completion criterion (7 checks) still pass unchanged, and the pytest suite covers both.

Assumptions

- $S_n = 200 \text{ nm/RIU}$. This is not given in the instructions; it is a typical value for a resonant dielectric sensor and only rescales Δn .
- Camera: $20 \mu\text{m}$ pixels, unit quantum efficiency and flat responsivity (W is not weighted), no dark current, no saturation modelled.
- Noise: 100 frames of $1 \times 10^7 \text{ e}^-$ with 100 e^- read noise per frame, independent between pixels. Shot noise is Poisson; read noise is Gaussian.
- The sensor is spatially uniform over the beam, so y is linear in the spectrum.
- The Stage 1 wavelength grid is reused as instructed. It was built for shifts 0 to +5 nm, so it is slightly asymmetric about 1550 nm; total power is stationary at 0 only to 6×10^{-5} .
- Pixel ranking and GLS weights use the noise at $\Delta\lambda_r = 0$ (the calibration point).

Limitations and next steps

- The fixed grating is a weak spectral coder: only about 2.5 grating periods are illuminated, so one spot radius corresponds to roughly 400 nm of wavelength. This limits both the bound and the gain over the A/B readout.
- The linear estimator is valid only within about $\pm 0.26y$ for all pixels; Gauss–Newton extends this when the noise is small, but for few pixels it needs bounds and becomes biased.
- Correlated noise, pixel crosstalk, detector nonlinearity and temperature drift of the resonance are not modelled. A greedy selector for correlated noise is implemented but not exercised.
- Next: make $\varphi(x, y)$ trainable and maximize $F(\varphi) = \sum_{j \in P_M} g_j^2 / \sigma_j^2$ with σ_j^2 depending on μ (Eq. 15), preferably using the amplitude-robust information so the design cannot win by exploiting total power.

Reproduce: `python src/embodiment2.py` (about 30 s) then `python src/make_e2_report.py`. Source: `src/spectral_coding.py` ($y = W s$), `src/estimator.py` (g , noise, estimators, Fisher ranking), `src/embodiment2.py` (deliverables and checks).